

TITLE PAGE

- **Problem Statement ID** –SIH26028
- **Problem Statement Title-** Dynamic Forecast of Expected Time of Arrival (ETA) for Coaching Trains
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name** - Pascals





From “Where is my train?” to “When will it actually arrive?”

PASSENGER PROBLEM

- Imagine Rahul is travelling overnight to Assam. His train is already 20 minutes late due to heavy rain and congestion. But he doesn't know whether the delay will become 30, 40, or 50 minutes.
- Current delay is not always enough, delay can grow due to congestion, weather and operations
- Repeated checking gives little visibility of what happens next
- Night travellers may wake too early- or miss their station

OUR SOLUTION

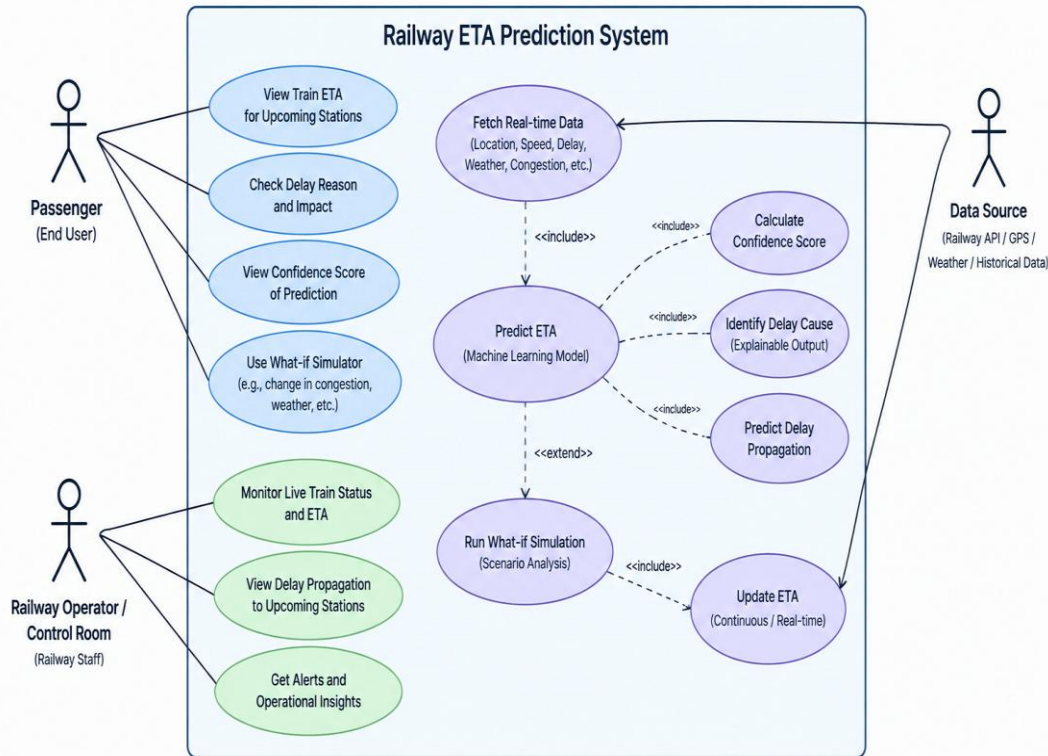
- If the passenger knows the updated ETA, they can plan accordingly and inform the person waiting at the destination about the expected arrival time.
- Continuously estimate future arrival time
- Combine live movement with route history and context
- Show dynamic ETA and expected delay
- Give confidence-aware alerts and ETA-aware alarm

INNOVATION & UNIQUENESS

- DynamicRail shows his updated ETA and the reason for the delay. If he is confused, he can ask the chatbot, and without network, he can check the last synced update offline.
- Predictive continuity- not only current-status tracking
- Route-wise Delay Trend and Confidence Score
- Explain why ETA changed
- Multilingual chatbot + offline sync



Real-Time Dynamic ETA Prediction for Coaching Trains



Simple System Flow

Input → Process → Predict → Inform

1

Collect live + historical data

Train movement, schedule, delay history and reliable context.

2

Prepare the data

Clean and combine the latest train movement with route information.

3

Predict future ETA

ML estimates arrival time, expected delay and confidence.

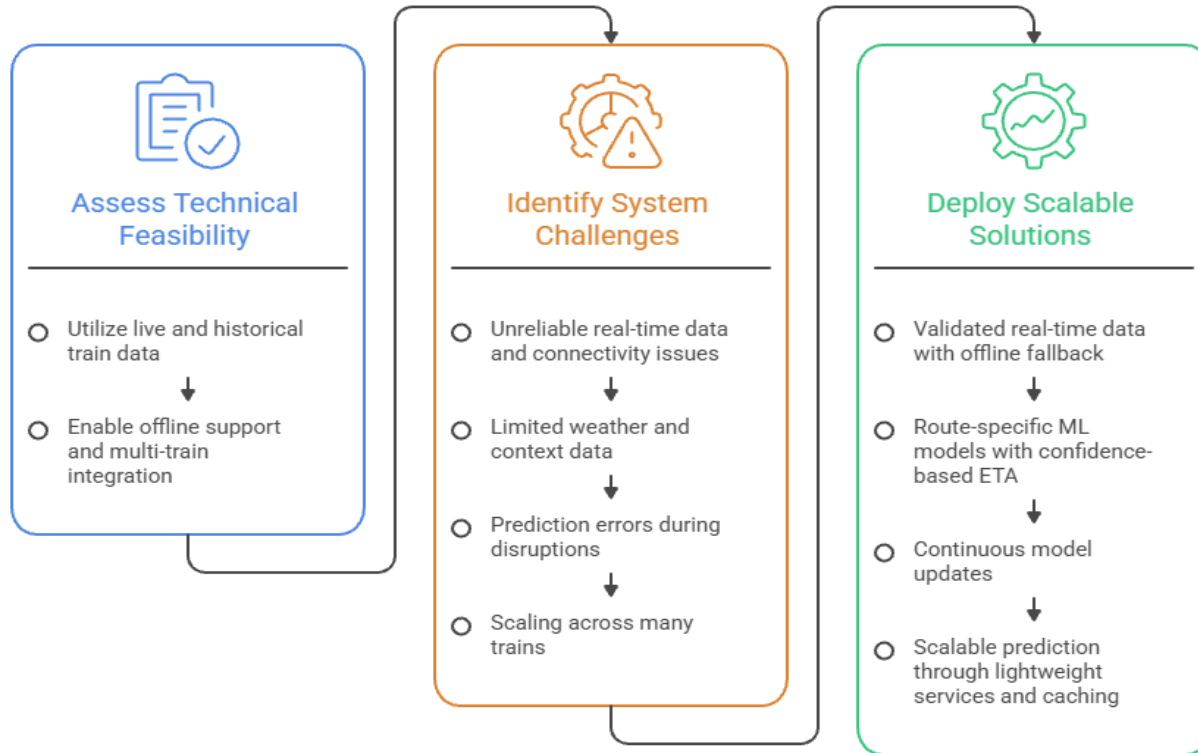
4

Show useful result

Passenger gets ETA, delay trend, reason and alert/alarm.

Key idea: the ETA keeps updating as new information arrives.

Implementing Modular ML-based ETA Prediction



Why This Idea Is Feasible

1. DATA

Use live train movement, schedules and historical route patterns. Add weather/context when reliable.

2. AI / ML

Train a model to learn delay patterns and estimate future arrival time with a confidence score.

3. PROTOTYPE

Start with selected routes, validate APIs/data and test route-level ETA performance.

4. RELIABILITY

If live data is weak, use the latest known information and show confidence instead of hiding uncertainty.

Implementation Plan

Phase 1

Validate data & APIs

Phase 2

Build and test ETA model

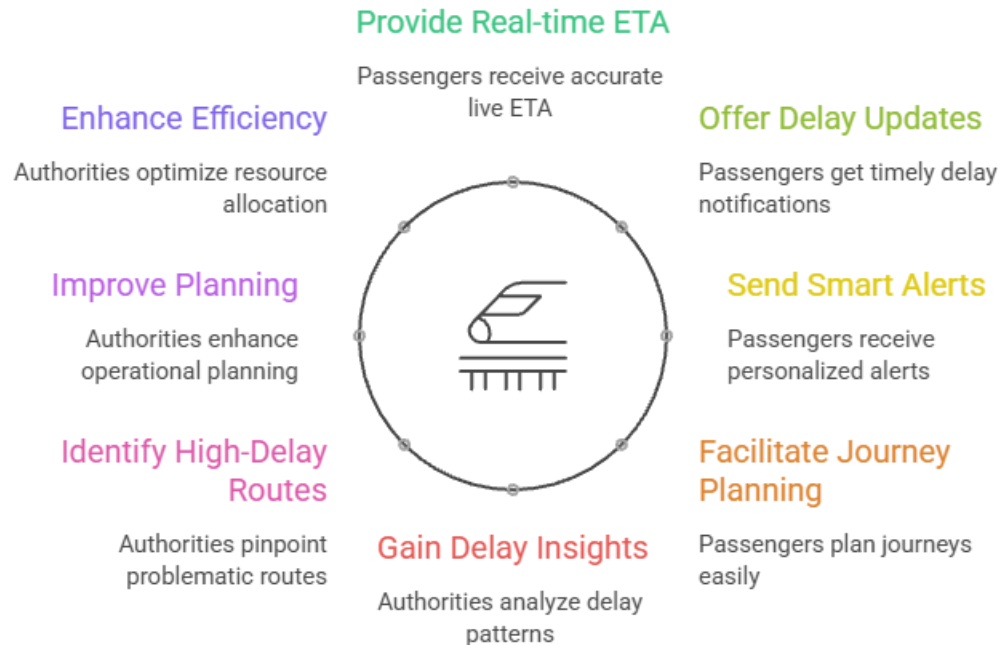
Phase 3

Connect alerts + user view

Made with  Notion

Better decisions for passengers; better visibility for railways Who Benefits?

Railway Service Improvement Cycle



Made with  Napkin

PASSENGERS

- Know the likely arrival time
- Plan station arrival better
- Reduce unnecessary waiting
- Lower risk of missing the station

RAILWAYS

- Track delay across the route
- Get better operational visibility
- Identify high-delay areas
- Support better planning

How We Measure Success

ETA accuracy

Compare predicted ETA with actual arrival.

Alert accuracy

Check whether alerts reach passengers at useful times.

User feedback

Measure whether passengers find ETA easier to use.

Future scope: more routes, stronger data coverage and better ETA accuracy after validation.

Existing status tools are the foundation; our innovation is predictive continuity

WHAT EXISTS

- NTES provides near-real-time train running information
- Arrival/departure status and schedules
- RailOne and other railway services support current movement tracking

IDENTIFIED GAP

- Current-status reporting answers: “Where is my train?”
- Our system adds: “How may delay evolve, and when is it likely to arrive?”

RESEARCH DIRECTION

- Delay Prediction:
https://www.sciencedirect.com/science/article/pii/S2210970622000166?utm_source=
- Speed and movement signals :
- Weather/context when reliable :
https://www.sciencedirect.com/science/article/abs/pii/S2210970624000180?utm_source=
- Confidence-aware ETA
- Passenger feedback loop

PROPOSED USER JOURNEY

Search

- Train number / journey



Track

- Live or last-known movement



Understand

- Delay trend + ETA window



Act

- Alert, alarm, route view

Reference: Press Information Bureau — National Train Enquiry System (NTES). Prototype phase: validate APIs, data licensing and route-level evaluation.